

$$X \text{ uma } f_{\theta}(x) = e^{\theta - x} \quad x > 0$$

$X_1, \dots, X_n$ uma X

Teste de hipótese α para $\begin{cases} H_0: \theta = \theta_0 \\ H_1: \theta = \theta_1 \end{cases}$

Leva N.P.

$$p(X_1, \dots, X_n) = \begin{cases} 1 & f_1^n > k f_0^n \\ \alpha & f_1^n = k f_0^n \\ 0 & f_1^n < k f_0^n \end{cases}$$

$$\int_{\theta}^n (x_1, \dots, x_n) = e^{n\theta - \sum x_i}$$

$$\underline{X: \sum, \theta} = e^{n\theta - \sum x_i}$$

$$I_{\left(\sum_{i=1}^n x_i > \theta\right)} (x_1, \dots, x_n)$$

$$\int_{\theta_1}^n (x_1, \dots, x_n) = e^{n\theta_1 - \sum x_i} \quad I_{\left(\sum_{i=1}^n x_i > \theta_1\right)} (x_1, \dots, x_n)$$

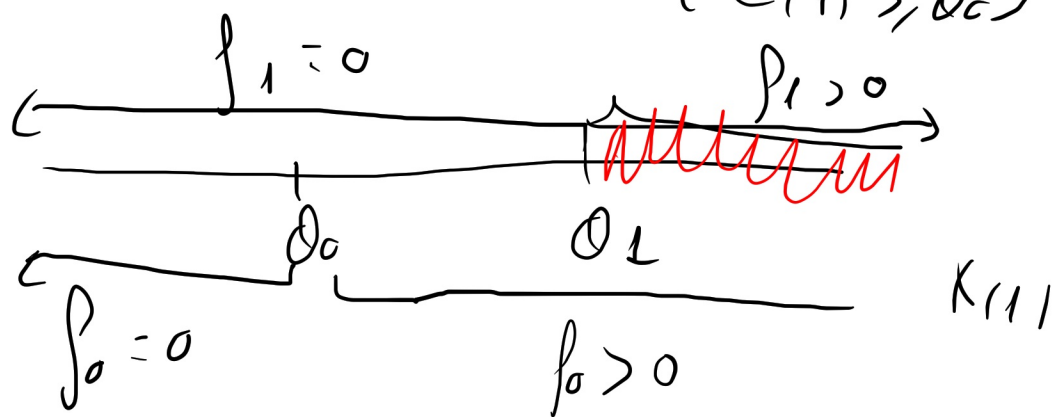
$$= \frac{e^{n\theta_1 - \sum x_i}}{e^{n\theta_0 - \sum x_i}}$$

$$= \begin{cases} e^{n\theta_1 - \sum x_i} & X_{(1)} > \theta_1 \\ 0 & X_{(1)} < \theta_1 \end{cases}$$

$$\int_{\theta_0}^n (x_1, \dots, x_n) = e^{n\theta_0 - \sum x_i} \quad I_{\left(\sum_{i=1}^n x_i > \theta_0\right)} (x_1, \dots, x_n)$$

$$= \begin{cases} e^{n\theta_0 - \sum x_i} & X_{(1)} > \theta_0 \\ 0 & X_{(1)} < \theta_0 \end{cases}$$

$\theta_0 < \theta_1$



$$\rho = \begin{cases} 1 & f_1^n / \rho_0^n > k \\ \alpha & f_1^n / \rho_0^n = k \\ 0 & f_1^n / \rho_0^n < k \end{cases}$$

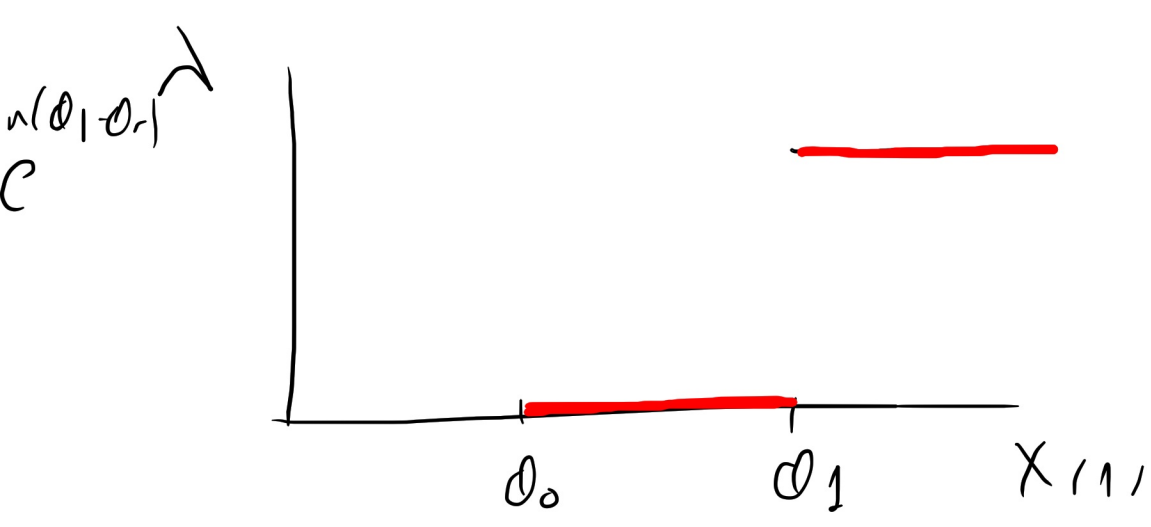
$$\frac{f_1^n}{\rho_0^n} = \lambda(x_1, \dots, x_n)$$

$$X_{(1)} > \theta_1 > \theta_0$$

$$\lambda(x_1, \dots, x_n) = \frac{e^{n\theta_1} e^{-\sum x_i}}{e^{n\theta_0} e^{-\sum x_i}} = e^{n(\theta_1 - \theta_0)}$$

$$\theta_0 \leq X_{(1)} < \theta_1$$

$$\lambda(x_1, \dots, x_n) = 0$$



$$\left\{ \begin{array}{l} k = e^{n(\theta_1 - \theta_0)} \\ k = 0 \end{array} \right.$$

$$k = 0$$

$$p = \begin{cases} 1 & \lambda > 0 \Leftrightarrow x_{(1)} \geq \theta_1 \\ \gamma & \lambda = 0 \Leftrightarrow \theta_0 \leq x_{(1)} < \theta_1 \\ 0 & \lambda < 0 \text{ nunca} \end{cases}$$

$$p = \begin{cases} 1 & x_{(1)} \geq \theta_1 \\ \gamma & \theta_0 \leq x_{(1)} < \theta_1 \end{cases}$$

$$\text{Tamaño} = \alpha = E_{\theta_0}(p) = 1 P_{\theta_0}(x_{(1)} \geq \theta_1) + \gamma P_{\theta_0}(\theta_0 \leq x_{(1)} < \theta_1)$$

$$\begin{aligned}
 P_{\theta_0}[\bar{X}_{(1)} > \theta_1] &= P_{\theta_0}[\bar{X}_1 > \theta_1, \dots, \bar{X}_n > \theta_1] \stackrel{\text{i.i.d. } \bar{X}}{=} \\
 &= \left(P_{\theta_0}[\bar{X} > \theta_1] \right)^n = \left(\int_{\theta_1}^{+\infty} e^{\theta_0 - x} dx \right)^n = \left(-e^{\theta_0 - x} \right)_{\theta_1}^{+\infty} = \\
 &= \left(0 - \left(-e^{\theta_0 - \theta_1} \right) \right)^n = e^{n(\theta_0 - \theta_1)}
 \end{aligned}$$

$$\alpha = e^{n(\theta_0 - \theta_1)} + \gamma \left(1 - e^{n(\theta_0 - \theta_1)} \right)$$

$$\gamma = \frac{\alpha - e^{n(\theta_0 - \theta_1)}}{1 - e^{n(\theta_0 - \theta_1)}} \leq 1 \Rightarrow \gamma \geq 0$$

$$\rho = \begin{cases} 1 & \bar{X}_{(1)} > \theta_1 \\ \frac{\alpha - e^{n(\theta_0 - \theta_1)}}{1 - e^{n(\theta_0 - \theta_1)}} & \theta_0 \leq \bar{X}_{(1)} < \theta_1 \end{cases}$$

$$\boxed{\alpha > e^{n(\theta_0 - \theta_1)}}$$

$$0 \leq \gamma \leq 1$$

$$K = e^{n(\theta_1 - \theta_0)}$$

$$0 \leq \alpha \leq 1$$

$$p = \begin{cases} 1 & \lambda > e^{n(\theta_1 - \theta_0)} \\ \gamma & \lambda = e^{n(\theta_1 - \theta_0)} \Leftrightarrow \bar{X}(1) \geq \theta_1 \\ 0 & \lambda < e^{n(\theta_1 - \theta_0)} \Leftrightarrow \theta_0 \leq \bar{X}(1) < \theta_1 \end{cases}$$

$$p = \begin{cases} \gamma & \bar{X}(1) \geq \theta_1 \\ 0 & \theta_0 \leq \bar{X}(1) < \theta_1 \end{cases} \quad \gamma = \frac{\alpha}{e^{n(\theta_1 - \theta_0)}} \begin{matrix} \geq 0 \\ \leq 1 \end{matrix}$$

$$\alpha = \text{bias} = E_{\theta_0}(p) = \gamma \underbrace{P_{\theta_0}(\bar{X}(1) \geq \theta_1)}_{e^{n(\theta_1 - \theta_0)}} + 0 \quad \left(\alpha \leq e^{n(\theta_1 - \theta_0)} \right)$$

Resumen: Para $\theta_0 < \theta_1$

(A)

$$\alpha \geq e^{n(\theta_0 - \theta_1)}$$

TUMP de tamaño α

$$\rho = \begin{cases} 1 & \bar{X}_{(1)} \geq \theta_1 \\ \frac{\alpha - e^{n(\theta_0 - \theta_1)}}{1 - e^{n(\theta_0 - \theta_1)}} & \theta_0 \leq \bar{X}_{(1)} < \theta_1 \end{cases}$$

(B)

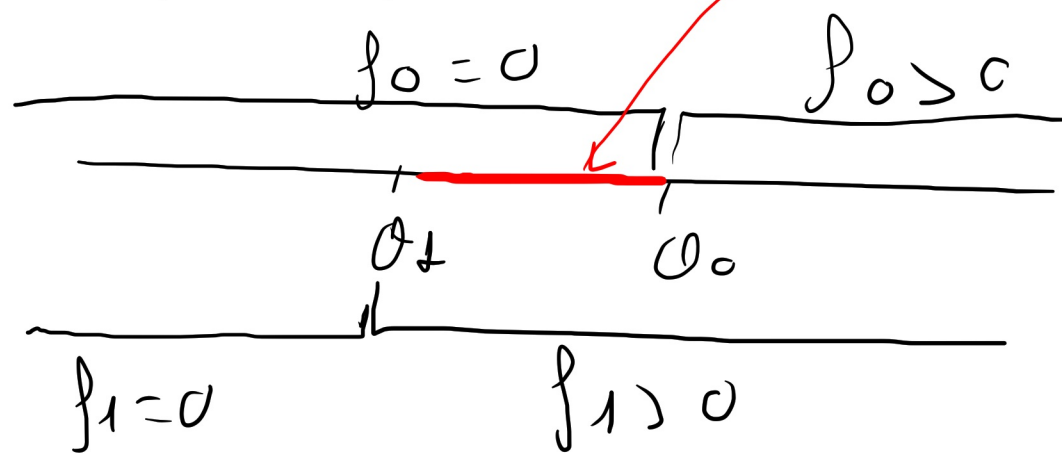
$$\alpha \leq e^{n(\theta_0 - \theta_1)}$$

TUMP de tamaño α

$$\rho = \begin{cases} \frac{\alpha}{e^{n(\theta_0 - \theta_1)}} & \bar{X}_{(1)} \geq \theta_1 \\ 0 & \theta_0 \leq \bar{X}_{(1)} < \theta_1 \end{cases}$$

Para $\theta_1 < \theta_0$

$f_1 > 0$ y $f_0 = 0$

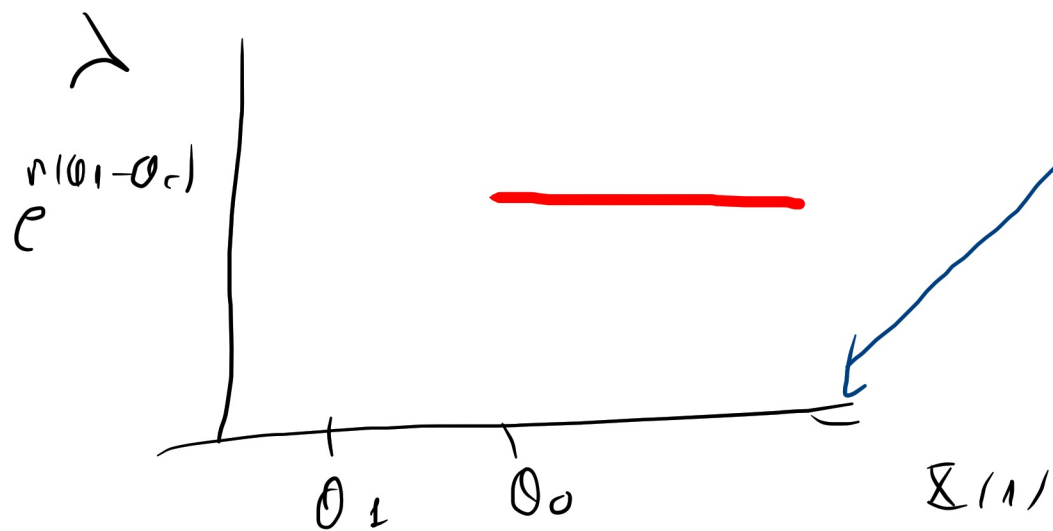


No puede dividirse por f_0
 luego en esa zona
 Rechazo

$$\rho = \begin{cases} 1 & \theta_1 < X_{(1)} < \theta_0 \\ 1 & \lambda > k \\ \gamma & \lambda = k \\ 0 & \lambda < k \end{cases}$$

$$\lambda = \frac{\int_1^n}{\int_0^n} = e^{n(\theta_1 - \theta_c)}$$

$\underbrace{X_{(1)} \geq \theta_0}_{\int_0 > 0}$



Sollc man plautes $k = e^{n(\theta_1 - \theta_c)}$

$$p = \begin{cases} 1 & \theta_1 \leq \bar{X}_{(1)} < \theta_0 \\ \lambda > e^{\frac{n(\theta_1 - \theta_c)}{\lambda}} & \text{never} \\ \gamma & \lambda = e^{\frac{n(\theta_1 - \theta_c)}{\lambda}} \Leftrightarrow \bar{X}_{(1)} \geq \theta_0 \\ 0 & \lambda < e^{\frac{n(\theta_1 - \theta_c)}{\lambda}} \text{ never} \end{cases}$$

$$p = \begin{cases} 1 & \theta_1 \leq \bar{X}_{(1)} < \theta_0 \\ \gamma & \bar{X}_{(1)} \geq \theta_0 \end{cases}$$

$$\alpha = \text{to max} = E_{\theta_c}(p) = \underbrace{P_{\theta_c}[\theta_1 \leq \bar{X}_{(1)} < \theta_0]}_{=0} + \gamma \underbrace{P_{\theta_0}[\bar{X}_{(1)} \geq \theta_0]}_{=1} = \gamma$$

Para $\alpha_1 < \alpha_0$

$$p = \begin{cases} 1 \\ \alpha \end{cases}$$

$$\alpha_1 \leq \bar{X}_{(1)} < \alpha_0$$

$$\bar{X}_{(1)} \geq \alpha_0$$

Para qualquer $\alpha_1 \leq \alpha_0$

+ LMP

P. teste: $\beta_p(\alpha_1) = E_{\alpha_1}(p) = P_{\alpha_1}[\alpha_1 \leq \bar{X}_{(1)} < \alpha_0] + \alpha P_{\alpha_1}[\bar{X}_{(1)} \geq \alpha_0] = x$

$$P_{\alpha_1}[\bar{X}_{(1)} \geq \alpha_0] = \left(P_{\alpha_1}[X \geq \alpha_0] \right)^n = \left(\int_{\alpha_0}^{+\infty} e^{-(x-\alpha_1)} dx \right)^n = \left(e^{-(\alpha_0-\alpha_1)} \right)^n = e^{-n(\alpha_0-\alpha_1)}$$

$$x = \left(1 - e^{-n(\alpha_0-\alpha_1)} \right) + \alpha e^{-n(\alpha_0-\alpha_1)}$$

$$\left(\int_{\alpha_0}^{+\infty} e^{-(x-\alpha_1)} dx \right)^n = 1$$

Para $\alpha_0 < \alpha_1$

(A) $\alpha \geq e^{n(\alpha_0-\alpha_1)}$

$$P_{\text{teste}} = E_{\alpha_1}(p) = 1 P_{\alpha_1}[\bar{X}_{(1)} \geq \alpha_1] + \frac{\alpha - e^{n(\alpha_0-\alpha_1)}}{1 - e^{n(\alpha_0-\alpha_1)}} P_{\alpha_1}[\alpha_1 \leq \bar{X}_{(1)} < \alpha_0]$$

$$= 1 //$$

(B) $\alpha \leq e^{n(\alpha_0-\alpha_1)}$

$$P_{\text{teste}} = E_{\alpha_1}(p) = \frac{\alpha}{e^{n(\alpha_0-\alpha_1)}} P_{\alpha_1}[\bar{X}_{(1)} \geq \alpha_1] = \frac{\alpha}{e^{n(\alpha_0-\alpha_1)}}$$

Test óptimo para n.s. α'

Si $\alpha_2 < \alpha_0$
$$p(\bar{x}_1, \dots, \bar{x}_n) = \begin{cases} 1 & \alpha_1 \leq \bar{x}_{(1)} < \alpha_0 \\ \alpha & \bar{x}_{(1)} \geq \alpha_0 \end{cases}$$

Potencia: $\beta_{\alpha}(\rho) = (1 - e^{n(\alpha_1 - \alpha_0)})^\alpha + \alpha e^{n(\alpha_1 - \alpha_0)}$ creciente en $\alpha = \text{tamaño}$

Como $\alpha \leq \alpha' \Rightarrow$ tomo $\alpha = \alpha'$ y así obtengo la potencia máxima.

Si $\alpha_0 < \alpha_1$ y $\alpha' \leq e^{n(\alpha_0 - \alpha_1)}$, como $\alpha \leq \alpha' \Rightarrow \alpha \leq e^{n(\alpha_0 - \alpha_1)}$ y el test más potente es (B)
$$p(\bar{x}_1, \dots, \bar{x}_n) = \begin{cases} \frac{\alpha}{e^{n(\alpha_0 - \alpha_1)}} & \bar{x}_{(1)} > \alpha_1 \\ 0 & \alpha_0 \leq \bar{x}_{(1)} < \alpha_1 \end{cases}$$

de potencia: $\beta_{\alpha}(\rho) = \frac{\alpha}{e^{n(\alpha_0 - \alpha_1)}}$ creciente en $\alpha \leq \alpha' \Rightarrow$ tomo tamaño $\alpha = \alpha'$ y así maximizo la potencia.

Si $\alpha_0 < \alpha_1$ y $\alpha' > e^{n(\alpha_0 - \alpha_1)}$, hay dos posibilidades

$$\frac{\alpha}{e^{n(\alpha_0 - \alpha_1)}} \quad \alpha'$$

Si tomo $\alpha \leq e^{n(\alpha_0 - \alpha_1)} \Rightarrow$ tomo el de antes y su potencia es $\alpha / e^{n(\alpha_0 - \alpha_1)}$

Si tomo $e^{n(\alpha_0 - \alpha_1)} < \alpha \leq \alpha' \Rightarrow$ tomo es (A)
$$p = \begin{cases} 1 & \bar{x}_{(1)} > \alpha_1 \\ \frac{\alpha - e^{n(\alpha_0 - \alpha_1)}}{1 - e^{n(\alpha_0 - \alpha_1)}} & \alpha_0 \leq \bar{x}_{(1)} < \alpha_1 \end{cases}$$
 de potencia 1

Luego, para maximizar la potencia me interesa $\alpha > e^{n(\alpha_0 - \alpha_1)}$ y para minimizar el tamaño $\alpha = e^{n(\alpha_0 - \alpha_1)}$.